

1 Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



In a normal home network, the internet comes from the ISP to the modem.

The modem changes the internet signal and sends it to the router.

The router shares the internet with all devices and creates Wi-Fi.

A switch connects more wired devices, such as a desktop PC.

An access point gives stronger Wi-Fi in large homes.

Devices like a laptop and smartphone connect wirelessly to the Wi-Fi.

2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Function	Example
Modem	Brings internet to your home.	Jio Fiber modem at home
Router	Shares internet and gives Wi-Fi.	Home Wi-Fi router
Switch	Connects many wired devices.	Office desktop computers
Access Point	Makes Wi-Fi stronger in a big area.	School or big house
Network Cable	Connects devices with a wire.	PC connected to router

3 Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1 Ethernet Cable



Use: Used to connect a PC, router, and switch in homes and offices

2 Coaxial Cable :



Use: Used for cable TV and broadband internet connections.

3 Fiber Optic Cable



Use: Used by internet providers for high-speed internet over long distances.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Point	Router	Switch
Main Use	Shares internet with devices	Connects wired devices together
Location	Home and office	Office, school, computer lab
Wi-Fi	Creates Wi-Fi	Does not create Wi-Fi
Data Handling	Sends data between different networks	Sends data inside the same network
Connection	Connects modem to devices	Connects PCs, printers, and servers
Ports	Usually 4–8 ports	Usually 8, 16, or 24 ports

